MA1522 Linear Algebra for Computing
L2 Lesson 12



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Semester 1, 2026/27

Row Space and Column Space of a Matrix

Let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{c}_1 & \boldsymbol{c}_2 & \cdots & \boldsymbol{c}_n \end{pmatrix} = \begin{pmatrix} \boldsymbol{r}_1 \\ \boldsymbol{r}_2 \\ \vdots \\ \boldsymbol{r}_m \end{pmatrix}$ be an $m \times n$ matrix, where

$\boldsymbol{c}_1, \boldsymbol{c}_2, \ldots, \boldsymbol{c}_n \in \mathbb{R}^m$ are the columns of $\boldsymbol{A}$ and $\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_m \in \mathbb{R}^n$ are the rows of $\boldsymbol{A}$.

The **row space** of $\boldsymbol{A}$, denoted $\mathrm{Row}(\boldsymbol{A})$, is the subspace of $\mathbb{R}^n$ spanned by its rows, i.e.

$$\mathrm{Row}(\boldsymbol{A}) = \mathrm{span}\{\boldsymbol{r}_1, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_m\}.$$

The **column space** of $\boldsymbol{A}$, denoted $\mathrm{Col}(\boldsymbol{A})$, is the subspace of $\mathbb{R}^m$ spanned by its columns, i.e.

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Theorem

Let $\boldsymbol{R}$ is a REF of a matrix $\boldsymbol{A}$. Then:

1. *the non-zero rows of $\boldsymbol{R}$ form a basis for $\mathrm{Row}(\boldsymbol{A})$.*
2. *the columns of $\boldsymbol{A}$ corresponding to the pivot columns of $\boldsymbol{R}$ form a basis for $\mathrm{Col}(\boldsymbol{A})$.*

Corollary

Let $\boldsymbol{A}$ be an $m \times n$ matrix. Then

$$\dim(\mathrm{Row}(\boldsymbol{A})) = \dim(\mathrm{Col}(\boldsymbol{A})).$$

We call this common dimension the **rank of $\boldsymbol{A}$**, *denoted* $\mathrm{rank}(\boldsymbol{A})$.

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We call this common dimension the **rank of $\boldsymbol{A}$**, *denoted* $\mathrm{rank}(\boldsymbol{A})$.

Question 1

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Let $\boldsymbol{A} = \begin{pmatrix} 1 & 2 & -1 & 0 & 3 \\ 0 & -1 & 3 & 1 & -1 \\ 1 & 0 & 5 & 2 & 1 \\ 3 & 1 & 12 & 5 & 4 \end{pmatrix}$.

Find a basis for $\mathrm{Row}(\boldsymbol{A})$ and a basis for $\mathrm{Col}(\boldsymbol{A})$, and write down $\mathrm{rank}(\boldsymbol{A})$. [Pg 10 and 16 of Chapter 4]

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Find a basis for $\mathrm{Row}(\boldsymbol{A})$ and a basis for $\mathrm{Col}(\boldsymbol{A})$, and write down $\mathrm{rank}(\boldsymbol{A})$. [Pg 10 and 16 of Chapter 4]

Solution.

$$\mathrm{RREF}(\boldsymbol{A}) = \begin{pmatrix} 1 & 0 & 5 & 2 & 1 \\ 0 & 1 & -3 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

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Solution.

$$\mathrm{RREF}(\boldsymbol{A}) = \begin{pmatrix} 1 & 0 & 5 & 2 & 1 \\ 0 & 1 & -3 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus $\{(1,0,5,2,1),(0,1,-3,-1,1)\}$ is a basis for $\mathrm{Row}(\boldsymbol{A})$ and $\{(1,0,1,3),(2,-1,0,1)\}$ is a basis for $\mathrm{Col}(\boldsymbol{A})$.

$\mathrm{rank}(\boldsymbol{A}) = 2$. □

Nullspace and Nullity

Let $\boldsymbol{A}$ be an $m \times n$ matrix. The **nullspace of $\boldsymbol{A}$**, denoted $\mathrm{Null}(\boldsymbol{A})$, is

$$\mathrm{Null}(\boldsymbol{A}) := \{\boldsymbol{x} \in \mathbb{R}^n \mid \boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}\}.$$

The **nullity of $\boldsymbol{A}$**, denoted $\mathrm{nullity}(\boldsymbol{A})$, is the dimension of $\mathrm{Null}(\boldsymbol{A})$, i.e.

$$\mathrm{nullity}(\boldsymbol{A}) := \dim(\mathrm{Null}(\boldsymbol{A})).$$

Theorem (Rank-nullity theorem)

$$\mathrm{rank}(\boldsymbol{A}) + \mathrm{nullity}(\boldsymbol{A}) = n.$$

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Question 2

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Let $\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 5 & 3 \\ 1 & -4 & -1 & -9 \\ -1 & 0 & -3 & 1 \\ 2 & 1 & 7 & 0 \\ 0 & 1 & 1 & 2 \end{pmatrix}$.

Find a basis for $\mathrm{Row}(\boldsymbol{A})$ and a basis for $\mathrm{Null}(\boldsymbol{A})$, and verify the rank-nullity theorem for $\boldsymbol{A}$.

Question 2: Solution

$$\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 5 & 3 \\ 1 & -4 & -1 & -9 \\ -1 & 0 & -3 & 1 \\ 2 & 1 & 7 & 0 \\ 0 & 1 & 1 & 2 \end{pmatrix} \xrightarrow{\text{Gauss-Jordan}} \begin{pmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Thus $\{(1, 0, 3, -1), (0, 1, 1, 2)\}$ is a basis for $\mathrm{Row}(\boldsymbol{A})$.

$$\begin{aligned} \mathrm{Null}(\boldsymbol{A}) &= \{(x_1, x_2, x_3, x_4) \mid x_1 + 3x_3 - x_4 = 0,\ x_2 + x_3 + 2x_4 = 0\} \\ &= \{(-3s + t, -s - 2t, s, t) \mid s, t \in \mathbb{R}\} \\ &= \{s(-3, -1, 1, 0) + t(1, -2, 0, 1) \mid s, t \in \mathbb{R}\} \\ &= \mathrm{span}\{(-3, -1, 1, 0), (1, -2, 0, 1)\}. \end{aligned}$$

Thus $\{(-3, -1, 1, 0), (1, -2, 0, 1)\}$ is a basis for $\mathrm{Null}(\boldsymbol{A})$.

$\mathrm{rank}(\boldsymbol{A}) = 2 = \mathrm{nullity}(\boldsymbol{A})$, and $\boldsymbol{A}$ has 4 columns. Since $\mathrm{rank}(\boldsymbol{A}) + \mathrm{nullity}(\boldsymbol{A}) = 4 =$ number of columns of $\boldsymbol{A}$, the rank-nullity theorem for $\boldsymbol{A}$ is verified.

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Theorem

Let $\boldsymbol{A}$ be an $m \times n$ matrix and let $\boldsymbol{B}$ be an $n \times p$ matrix. Then

$$\mathrm{Col}(\boldsymbol{AB}) \subseteq \mathrm{Col}(\boldsymbol{A})$$

and

$$\mathrm{rank}(\boldsymbol{AB}) \leq \min(\mathrm{rank}(\boldsymbol{A}), \mathrm{rank}(\boldsymbol{B})).$$

Question 3

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Let $\boldsymbol{A}$ be an $m \times n$ matrix and let $\boldsymbol{B}$ be an $n \times p$ matrix. Which of the following statements is/are (always) true?

a. $\text{Row}(\boldsymbol{AB}) \subseteq \text{Row}(\boldsymbol{A})$.

b. $\text{Row}(\boldsymbol{AB}) \subseteq \text{Row}(\boldsymbol{B})$.

c. $\text{Null}(\boldsymbol{B}) \subseteq \text{Null}(\boldsymbol{AB})$.

d. $\text{nullity}(\boldsymbol{AB}) \geq \max(\text{nullity}(\boldsymbol{A}), \text{nullity}(\boldsymbol{B}))$.



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a. $\text{Row}(\boldsymbol{AB}) \subseteq \text{Row}(\boldsymbol{A})$. ..... **Ans: False**

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d. $\text{nullity}(\boldsymbol{AB}) \geq \max(\text{nullity}(\boldsymbol{A}), \text{nullity}(\boldsymbol{B}))$. ..... **Ans: False**



Question 3: Solution

a. $\mathrm{Row}(\boldsymbol{AB}) \subseteq \mathbb{R}^p$ and $\mathrm{Row}(\boldsymbol{A}) \subseteq \mathbb{R}^n$. So if $n \neq p$, then $\mathrm{Row}(\boldsymbol{AB}) \not\subseteq \mathrm{Row}(\boldsymbol{A})$.

b. $\mathrm{Row}(\boldsymbol{AB}) = \mathrm{Col}((\boldsymbol{AB})^T) = \mathrm{Col}(\boldsymbol{B}^T\boldsymbol{A}^T) \subseteq \mathrm{Col}(\boldsymbol{B}^T) = \mathrm{Row}(\boldsymbol{B})$.

c. If $\boldsymbol{v} \in \mathrm{Null}(\boldsymbol{B})$, then $\boldsymbol{Bv} = \boldsymbol{0}$, and so $(\boldsymbol{AB})\boldsymbol{v} = \boldsymbol{A}(\boldsymbol{Bv}) = \boldsymbol{A0} = \boldsymbol{0}$. Thus $\boldsymbol{v} \in \mathrm{Null}(\boldsymbol{AB})$.

d. Let $\boldsymbol{A} = \begin{pmatrix} 1 & 0 \end{pmatrix}$ and $\boldsymbol{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. Then $\boldsymbol{AB} = \begin{pmatrix} 1 \end{pmatrix}$.

Thus $\mathrm{Null}(\boldsymbol{A}) = \{\begin{pmatrix} 0 \\ t \end{pmatrix} \mid t \in \mathbb{R}\}$ and $\mathrm{Null}(\boldsymbol{AB}) = \{0\}$. Hence

$$\mathrm{nullity}(\boldsymbol{A}) = 1 > 0 = \mathrm{nullity}(\boldsymbol{AB}).$$

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$$\mathrm{nullity}(\boldsymbol{A}) = 1 > 0 = \mathrm{nullity}(\boldsymbol{AB}).$$

Next Lesson ...

We will explore **Sections 5.1 and 5.2**.
Please watch/review the video clips for these sections before then.